# Introduction to Operations Research, Deterministic Models

## Lecture 2: Introduction of Linear Programming

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# Recap: LP model

**General form of LP:**

$$
\begin{aligned}
\text{maximize} \quad & Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n \\
\text{subject to} \quad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1 \\
& a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2 \\
& \quad \vdots \\
& a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m \\
& x_1 \geq 0, x_2 \geq 0, \ldots x_n \geq 0
\end{aligned}
$$

# Recap: solutions to the LP model

- feasible solution: a solution for which all the constraints are satisfied
- infeasible solution: a solution for which at least one constraint is violated
- optimal solution: a feasible solution that has the largest value of the objective function that is to be maximized

# Recap: solutions to the LP model

- feasible solution: a solution for which all the constraints are satisfied
- infeasible solution: a solution for which at least one constraint is violated
- optimal solution: a feasible solution that has the largest value of the objective function that is to be maximized
- There could be no feasible solutions, or unbounded objective, which lead to no optimal solutions
- There could also be multiple optimal solutions

# Recap: assumptions of LP

## Proportionality

The contribution of each activity to the value of the objective/constraint function is proportional to the level of the activity $x_j$.

## Additivity

Every function in a linear programming model (whether the objective function or the constraint functions) is the sum of the individual contributions of the respective activities.

## Divisibility

Decision variables are allowed to have any values, including fractional, noninteger values.

## Certainty

The value assigned to each parameter of a LP model is assumed to be a known constant.

# LP: more applications

# Application: capital budgeting problem

Compare two investments:

- Investment 1: -10,000 at time 0; 24,000 at time 1; -14,000 at time 2
- Investment 2: -6,000 at time 0, 8,000 at time 1, -1,000 at time 2

- If there is no annual interest:
  - net present value of Investment 1: -10000 + 24000 - 14000 = 0
  - net present value of Investment 2: -6000 + 8000 -1000 = 1000

- If the annual interest is $20\%$:
  - net present value of Investment 1: $-10000 + \frac{24000}{1+0.2} - \frac{14000}{(1+0.2)^2} = 277.78$
  - net present value of Investment 2: $-6000 + \frac{8000}{1+0.2} - \frac{1000}{(1+0.2)^2} = -27.78$

**Net present value** (NPV) is an important concept to compare the desirability of different investments.

# Application: capital budgeting problem

Star Oil Company is considering five different investment opportunities. The cash outflows in time 0 and time 1, and NPV (in millions of dollars) are given in the following table.

| | Inv 1 | Inv 2 | Inv 3 | Inv 4 | Inv 5 |
|---|---|---|---|---|---|
| Time 0 cash outflow | 11 | 53 | 5 | 5 | 29 |
| Time 1 cash outflow | 3 | 6 | 5 | 1 | 34 |
| NPV | 13 | 16 | 16 | 14 | 39 |

Star Oil Company has \$40 millions available for investments now (time 0); it estimates that one year from now (time 1) \$20 million will be available for investment.

Star Oil Company may purchase any fraction of the investment, and suppose that the cash outflows and NPV are adjusted accordingly.

The company wants to maximize the NPV that can be obtained by investing the five investments. Formulate an LP to help achieve the goal.

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What assumptions may not be satisfied in reality?

# Application: blending problem

Blending problem: various inputs must be blended in some desired proportion to produce goods for sale which are often amenable to LP analysis.

For instance,

- Blending various types of crude oils to produce different types of gasoline and other outputs
- Blending various types of chemicals to produce other chemicals
- Blending various livestock feeds in an attempt to produce a minimum-cost feed mixture
- Mixing various ores to obtain ore of a specified quality
- Mixing various types of papers to produce recycled paper of varying quality

# Application: blending problem

Sunco Oil manufactures three types of gasoline. Each type is produced by blending three types of crude oil. Sunco can purchase up to 5000 barrels of each type of crude oil daily.

| Gas | price per barrel | Crude | price per barrel |
|---|---|---|---|
| 1 | 70 | 1 | 45 |
| 2 | 60 | 2 | 35 |
| 3 | 50 | 3 | 25 |

| Crude | octane rating | sulfur |
|---|---|---|
| 1 | 12 | 0.5 |
| 2 | 6 | 2.0 |
| 3 | 8 | 3.0 |

Gas 1 must have an average octane rating of at least 10 and contain at most 1% sulfur;
Gas 2 must have an average octane rating of at least 8 and contain at most 2% sulfur
Gas 3 must have an average octane rating of at least 6 and contain at most 1% sulfur

• It costs \$4 to transform one barrel of oil to one barrel of gas, and Sunco can produce up to 14,000 barrels of gasoline daily.

• Sunco's customers require the following amounts of each gasoline: 3000 barrels of gas 1 per day; 2000 barrels of gas 2 per day; 1000 barrels of gas 3 per day.

• To stimulate demand for its products, each dollar spend daily in advertising a particular type of gas increases its daily demand by 10 barrels. What are the best blending and advertising plan to maximize the daily profits?

# Application: blending problem

**Decision variables:**
$a_i =$ dollars spent daily on advertising gas $i$; $x_{ij} =$ barrels of crude oil $i$ to produce gas $j$ daily

**Assumptions:**
- The gasoline must be sold on the day it is produced;
- the quality level of a mixture is a linear combination of each input used in the mixture;
- the demand increases proportionally with the advertisement investment.

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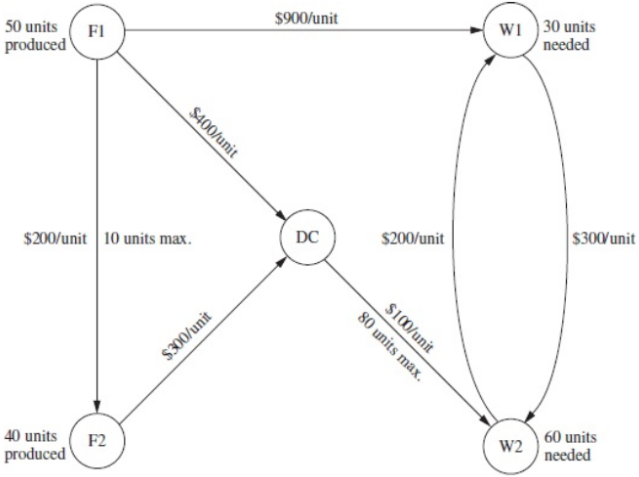
The Chevron Corporation is one of the world's leading integrated energy companies. Chevron developed the earliest applications in LP such as the blending problem. A typical refinery might have 20 different components to be blended into four or more grades of gasoline differing in octane and other properties by marketing area. The estimated cumulative value to Chevron now approaches **$ 1 billion annually**. In recognition of this and related work, INFORMS awarded Chevron the prestigious 2015 INFORMS prize.

# Application: distribution network problem

A distribution company will produce a new product at two different factories, and then the product must be shipped to two warehouses, where either factory can supply either warehouse.




- F1 can ship directly to W1; F1 can ship to W2 via three routes: F1 $\rightarrow$ DC $\rightarrow$ W2; F1 $\rightarrow$ W1 $\rightarrow$ W2; F1 $\rightarrow$ F2 $\rightarrow$ DC $\rightarrow$ W2.
- F2 has just one route to W1 (F2 $\rightarrow$ DC $\rightarrow$ W2 $\rightarrow$ W1) and one route to W2 (F2 $\rightarrow$ DC $\rightarrow$ W2).
- Except (F1 $\rightarrow$ F2) and (DC $\rightarrow$ W2), other lanes have sufficient shipping capacity to handle everything theses factories can send.
- How much to ship through each shipping lane with the minimal shipping cost?

# Application: distribution network problem

**Decision variables:**

Let $x_{\mathrm{F1-F2}}$, $x_{\mathrm{F1-DC}}, x_{\mathrm{F1-W1}}, x_{\mathrm{F2-DC}}, x_{\mathrm{DC-W2}}, x_{\mathrm{W1-W2}}, x_{\mathrm{W2-W1}}$ represent the amounts shipped through the seven respective lanes.

**Net flow constraint:**

$\boxed{\text{amount shipped out - amount shipped in} = \text{required amount}}$ for each location

This type of problems is called the **minimum cost flow problem**

# Simplex method for LP: geometric view

# Geometric concepts

The Simplex method is an algebraic procedure. To provide the intuitions, we introduce the Simplex from the geometric viewpoint.

## Wyndor problem

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\text{maximize} \quad & Z = 3x_1 + 5x_2 \\
\text{subject to} \quad & x_1 \leq 4 \\
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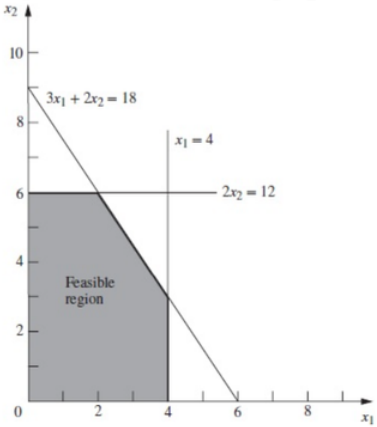
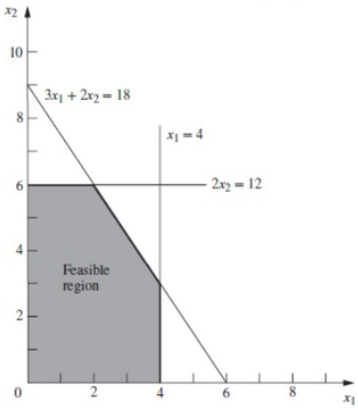


# Geometric concepts




For an LP problem with $n$ decision variables,

- each **constraint boundary** is a line that forms the boundary of what is permitted by the corresponding constraint.

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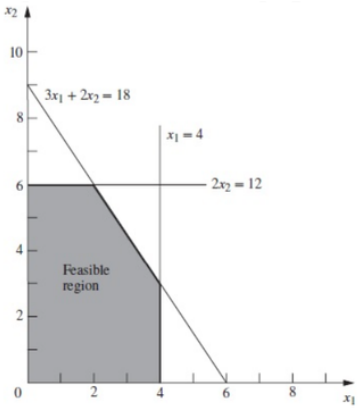


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- each **constraint boundary** is a line that forms the boundary of what is permitted by the corresponding constraint.
- each of its **corner-point solutions** lies at the intersection of $n$ constraint boundaries. some of which within the feasible region are **corner-point feasible** (CPF) solutions, and others are corner-point infeasible solutions.

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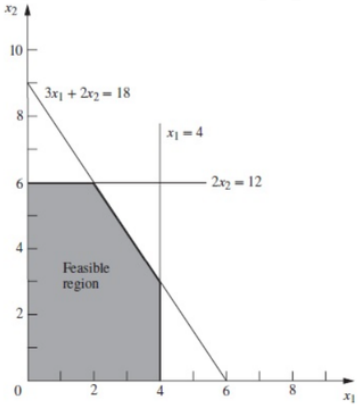


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- two CPF solutions are **adjacent** to each other if they share $n-1$ constraint boundaries. The two adjacent CPF solutions are connected by a line segment that lies on these same shared constraint boundaries. Such a line segment is referred to as an **edge** of the feasible region.

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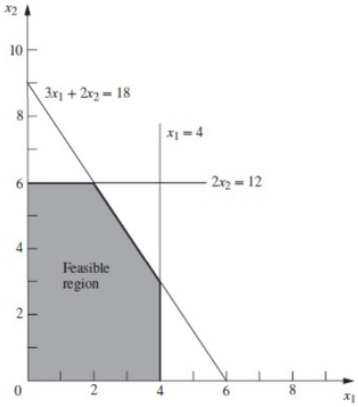


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• two CPF solutions are **adjacent** to each other if they share $n-1$ constraint boundaries. The two adjacent CPF solutions are connected by a line segment that lies on these same shared constraint boundaries. Such a line segment is referred to as an **edge** of the feasible region.

**Optimality test:** For an LP problem with at least one optimal solution, if a CPF solution has no adjacent CPF solutions that are better, then it must be an optimal solution.

# Simplex method: geometric outline

**Initialization:** choose $(0, 0)$ as the initial CPF solution to examine.

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**Optimality test:** find the two CPF solutions (0,6) and (4,0) that are adjacent to $(0, 0)$;

1. Choose the edge to move along with
2. Stop at the first new constraint boundary
3. Compute the intersection of the new set of constraint boundaries
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**Iteration 1:** move to a better adjacent CPF solution, say (0,6).

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**Iteration 1:** move to a better adjacent CPF solution, say (0,6).

**Optimality test:** the CPF solution (2,6) that adjacent to (0,6) obtains better objective value, thus (0,6) is not an optimal solution.

**Iteration 2:** move to a better adjacent CPF solution (2,6).

**Optimality test:** None of adjacent CPF of (2,6) achieves the better objective value, thus we conclude that (2,6) is an optimal solution.

# Key solution concepts

**Solution concept 1:** The simplex method focuses solely on CPF solutions. For any problem with at least one optimal solution, finding one requires only finding a best CPF solution.

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**Solution concept 3:** Whenever possible, the initialization of the simplex method chooses the origin to be the initial CPF solution.

**Solution concept 4:** Given a CPF solution, it is much quicker computationally to gather information about its adjacent CPF solutions than about other CPF solutions. Consequently, the entire path followed by the Simplex method is along the edges of the feasible region until eventually reaching an optimal solution.

# Key solution concepts

**Solution concept 5:** Each of the edges that emanate from the current CPF solution leads to an adjacent CPF solution. A positive rate of improvement in the objective value that would be obtained by moving along the edge implies that the adjacent CPF solution is better than the current CPF solution, whereas a negative rate of improvement in $Z$ implies that the adjacent CPF solution is worse.

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**Solution concept 5:** Each of the edges that emanate from the current CPF solution leads to an adjacent CPF solution. A positive rate of improvement in the objective value that would be obtained by moving along the edge implies that the adjacent CPF solution is better than the current CPF solution, whereas a negative rate of improvement in $Z$ implies that the adjacent CPF solution is worse.

**Solution concept 6:** Instead of solving all adjacent CPF solutions, the simplex method simply identifies the **rate of improvement** in the objective value that would be obtained by moving along the edge. It then chooses to move along the one with **the largest positive rate of improvement** in $Z$. If there is no positive rate of improvements, then the current CPF solution is optimal.

# Algebraic concepts in Simplex methods

# Algebraic concepts

We consider the Wyndor problem again.

**General form** of the model

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \leq 4 \\ & 2x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1 \geq 0, x_2 \geq 0 \end{array}$$

**Augmented form** of the model

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad\quad + x_3 = 4 \\ & \qquad 2x_2 \quad + x_4 = 12 \\ & 3x_1 + 2x_2 \qquad\quad + x_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{array}$$

- $x_3, x_4, x_5$ are referred to as *slack variables* which are used to transform $\leq$ inequality into equation with non-negativity constraints.

- An **augmented solution** is a solution for the original variables that has been augmented by the corresponding values of the slack variables. e.g. ( 3, 2, ?, ?, ? )

# Algebraic concepts

**Augmented form** of the Wyndor model

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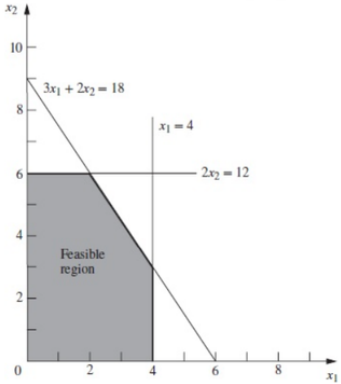
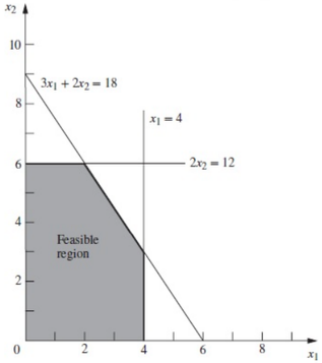
\end{array}
$$




- For an augmented solution, if the slack variable
  - equals 0: the solution lies on the corresponding constraint boundary;
  - greater than 0: the solution lies on the feasible side of this constraint boundary;
  - less than 0: the solution lies on the infeasible side of this constraint boundary

# Algebraic concepts

- a basic solution is an augmented corner-point solution; e.g. (4, 6, 0, 0, -6)
- a basic feasible (BF) solution is an augmented corner-point feasible (CPF) solution; e.g. (0, 6, 4, 0, 6)
- the only difference between basic solutions and corner-point solutions (between BF and CPF solutions) is whether the values of the slack variables are included.